



PHENIKAA UNIVERSITY
Faculty of Computer Science

CSE703016: Introduction to HCI

Lecture 09: Sketching

Instructor: Dr. Mai Xuan Trang

Today's Topic

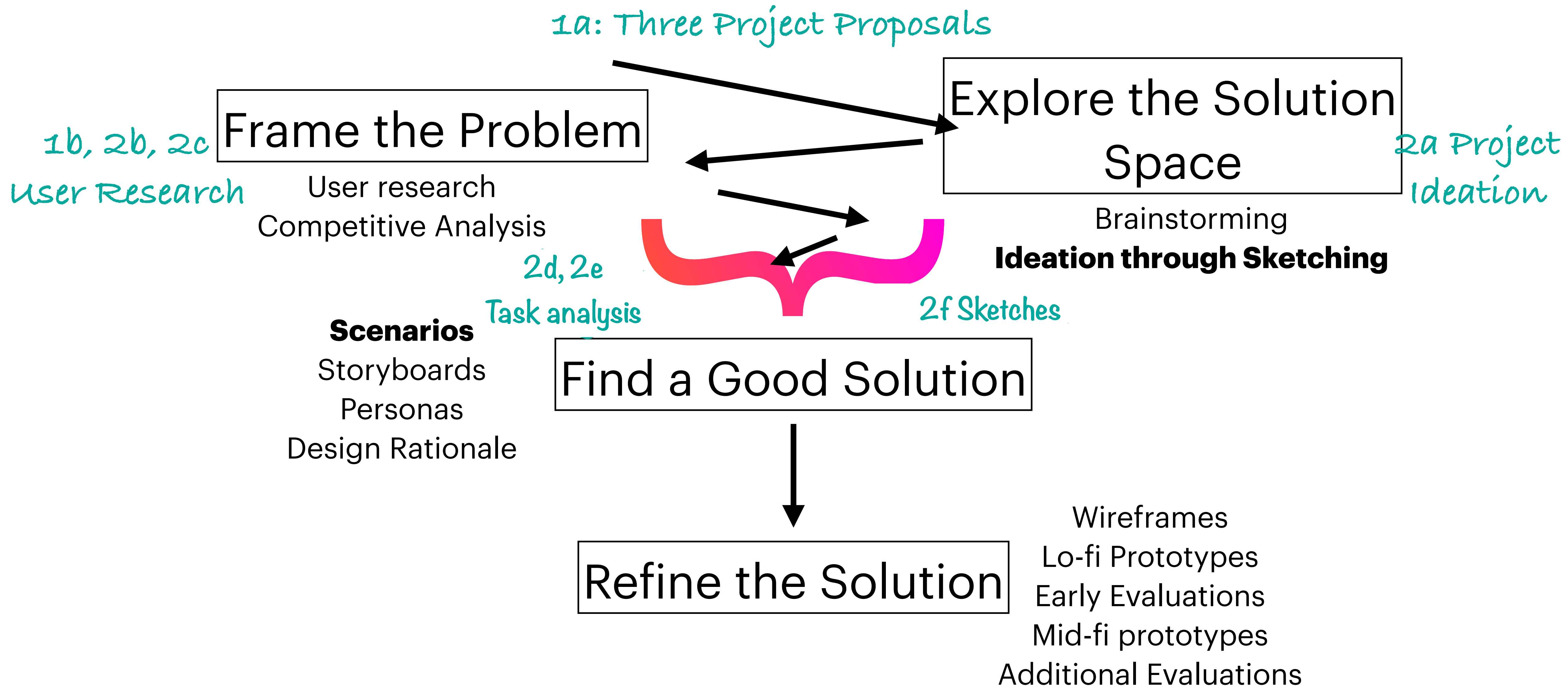
- Where are we now?
- Sketching
- Rapid Prototyping
- Teamwork time on 2d, 2e

Where are we now?

Looking back...

- Lectures finished:
 - Design Process: design diamond, iteration, ideation, critique
 - Design Principles: learnability, safety, efficiency
 - User Research: methods, best practices and tips, task analysis
- Now:
 - **Prototyping and User Testing:** the bulk of the class (6 lectures)!

Where are we now?



Sketching

2a Sketching

Movies

Theater: Shattuck Cinemas
Phone: (510) 665-1342 Dist: 1.5 mi
Address: 2122 Shattuck Ave
Berkeley, 94709
Cost: \$8.50 normal, \$6.00 senior, \$4.50 matinee

Map-It

<u>Art of War</u>	☆☆☆
(10:00)-(1:00)-4:00-7:00-10:00	
<u>Bittersweet Motel</u>	☆☆☆☆
(11:00)-(1:30)-4:00-6:30-9:00	
<u>Godzilla</u>	☆☆
(10:30)-(2:00)-5:30-9:00	
<u>The Cell</u>	☆☆☆☆
(11:00)-(1:00)-3:00-5:00-7:00-9:00	

STORE FOR THE STYLE-CHALLENGED

As is...

...  ...

...  ...

...  ...



As it should be...

outfit#1



outfit#2

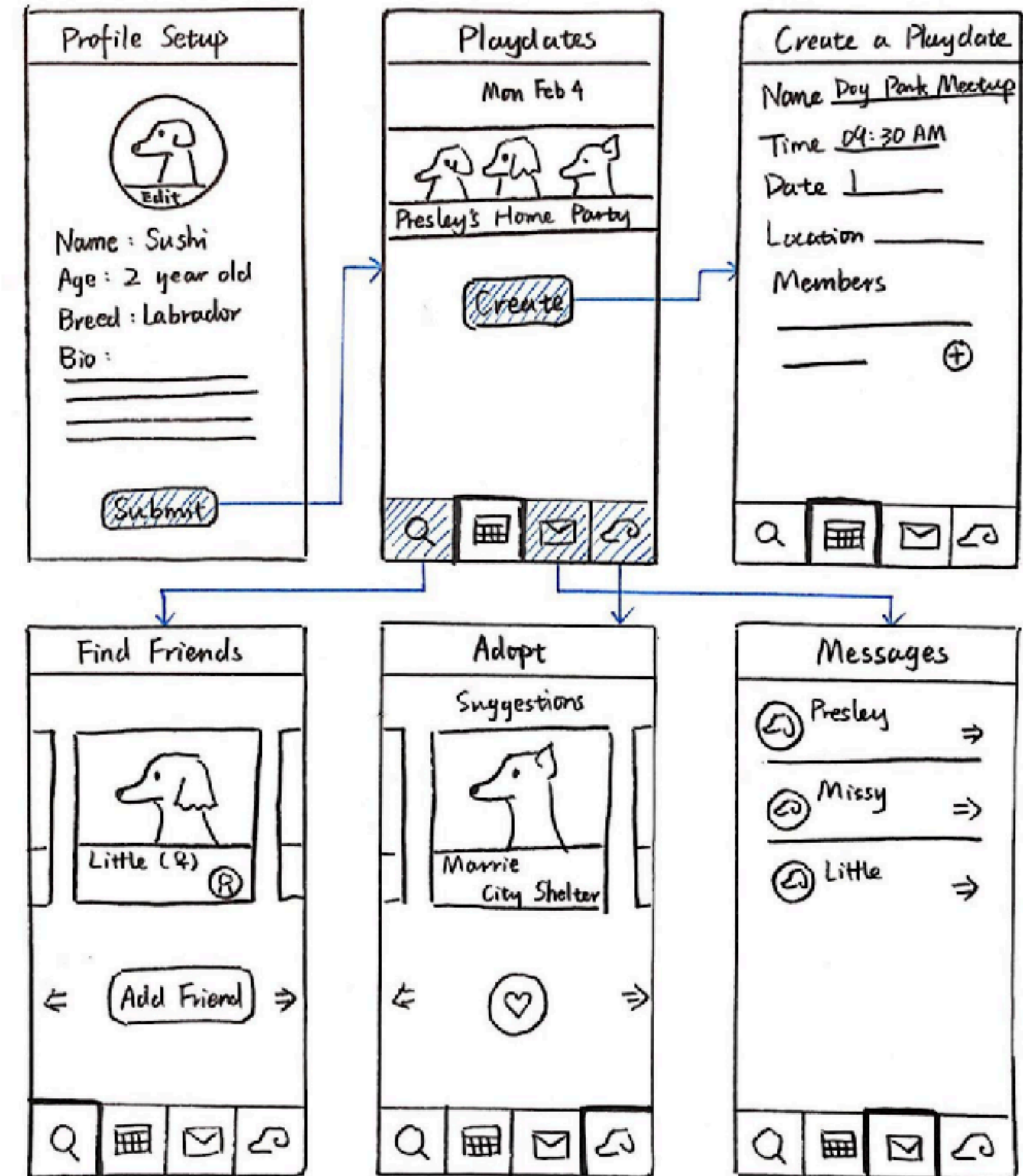
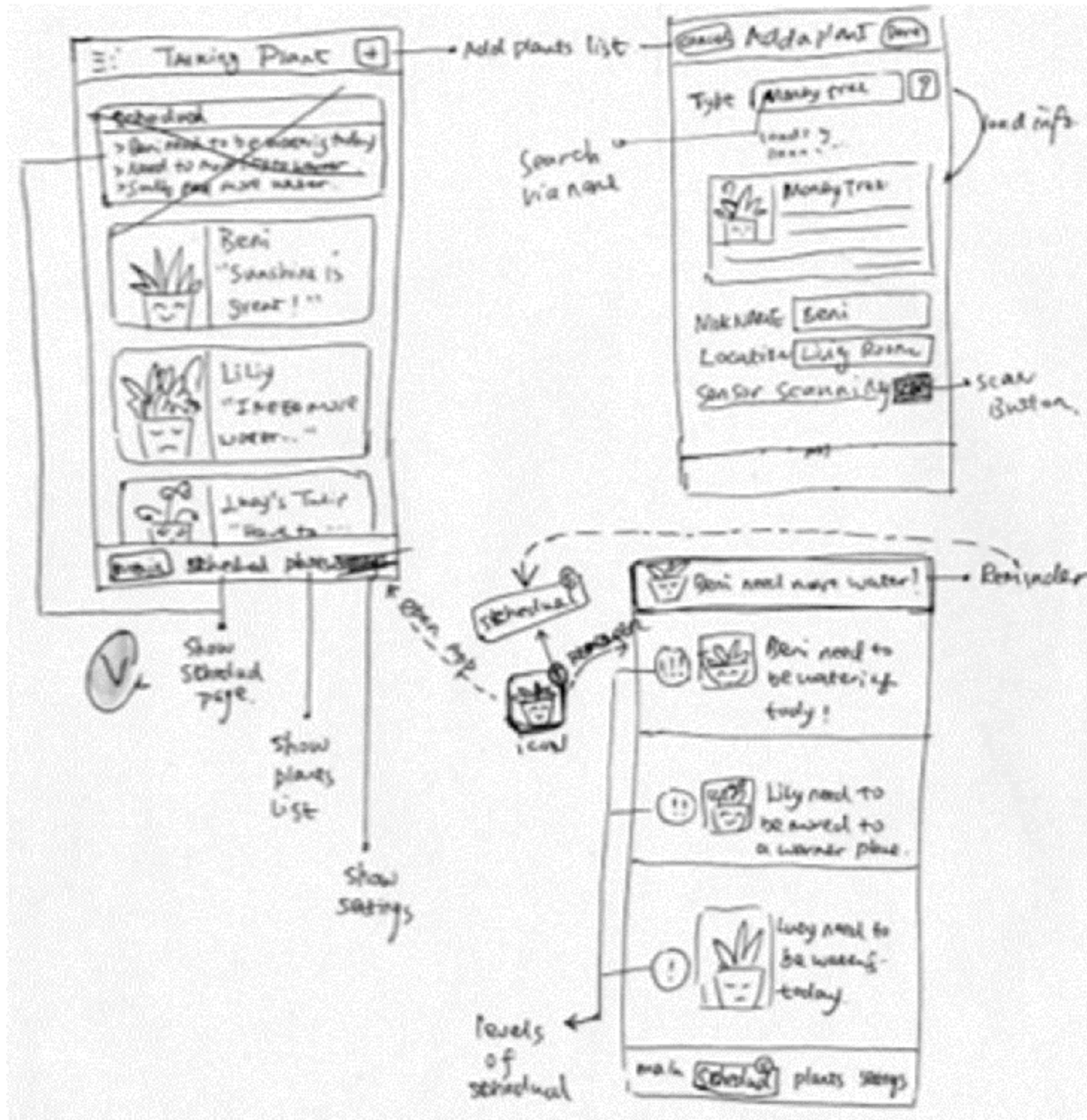


outfit#3



(pre-selected to match so you don't have to choose.)

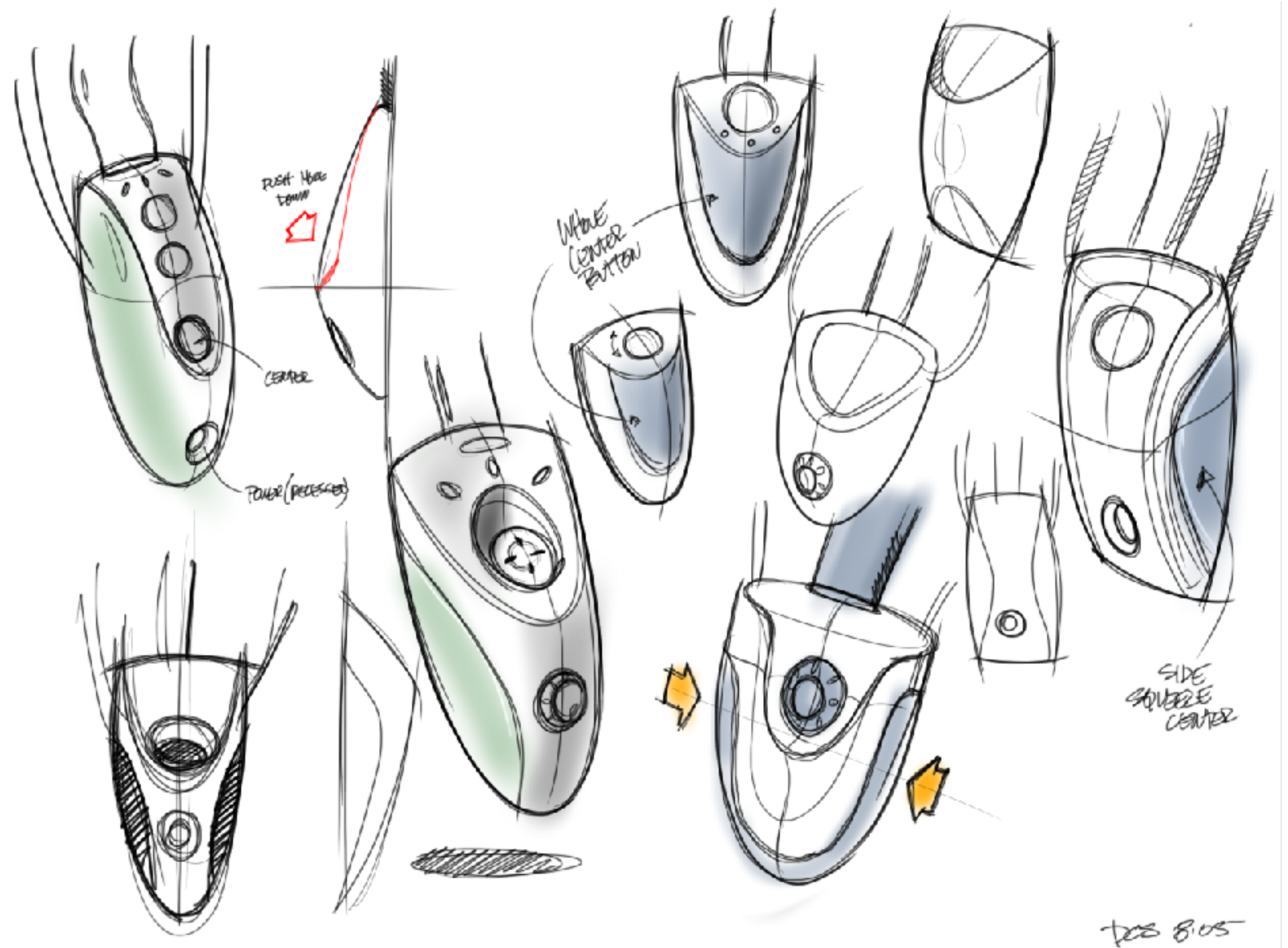
2f Sketching



Why Sketch?

- Sketching is the fastest instance of design iteration (an entire design-implement-evaluate cycle in as little as a few minutes!)
- We are still in the tightest part of the spiral in the **spiral model**
- Because it's so cheap, we can also **parallel prototype** (which you've learned is better for design)
- All these things help us **boost creativity!**
- As our ideas get more in-depth (moving towards higher fidelity prototypes), we narrow and switch to **serial prototyping**

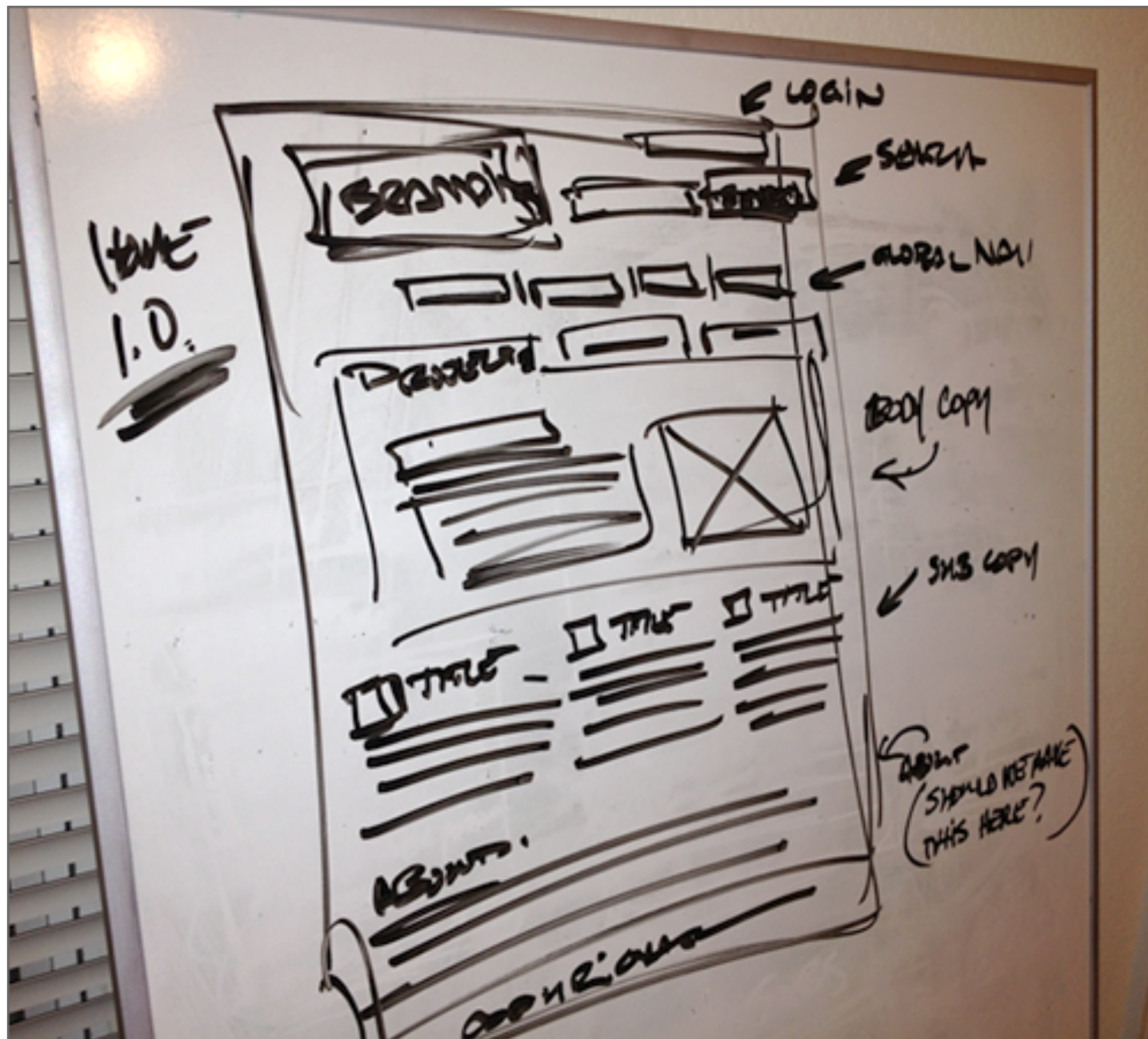
Sketching is...
A **process** that enables you to
think through ideas and
convey design ideas to others
very early in the design phase



Properties of Sketches

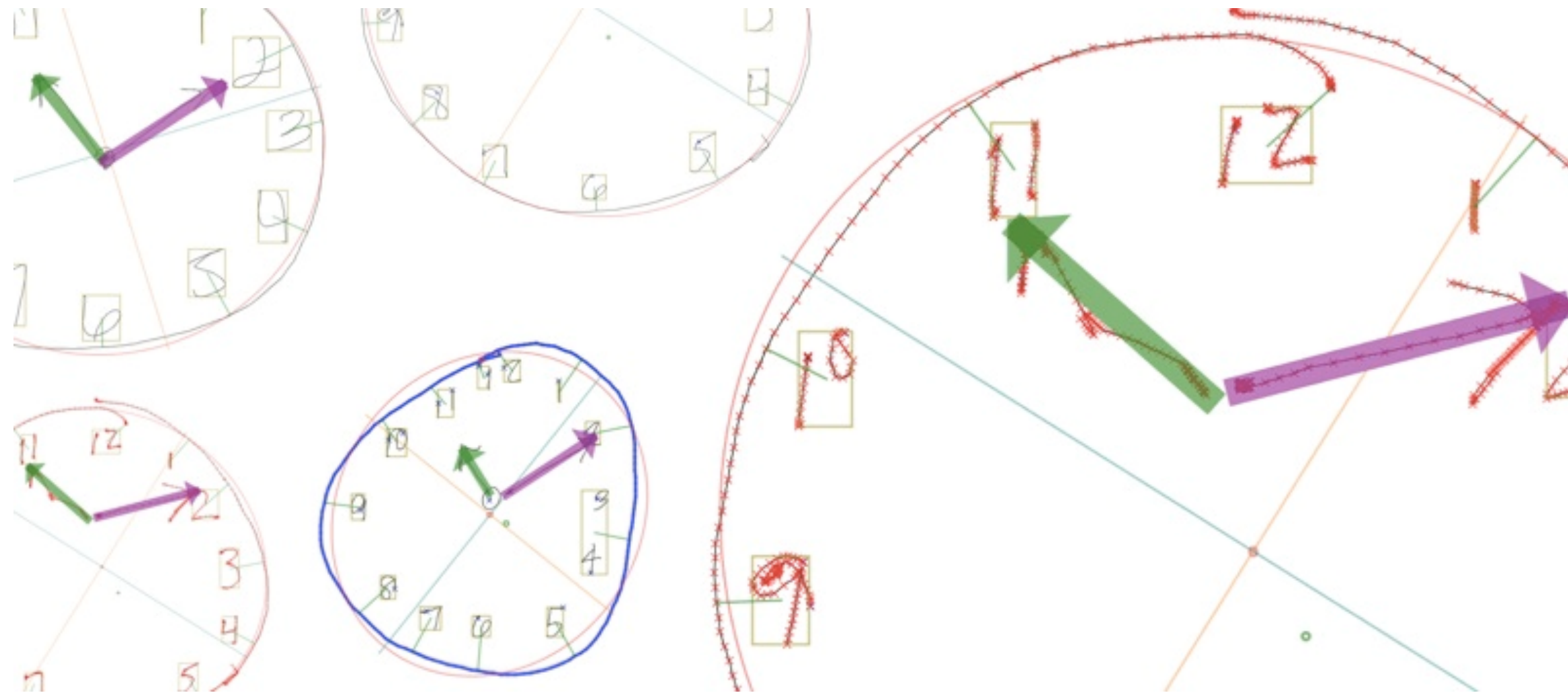
- Quick
- Timely
- Inexpensive
- Disposable
- Plentiful
- Clear Vocabulary
- Distinct Gesture
- Minimal Detail
- Appropriate Refinement
- Suggest and Explore
- Ambiguous

Quick



A sketch is quick to make,
or at least gives that impression

Timely



A sketch can be provided
when needed

Inexpensive



Cost must not inhibit the ability to explore a concept, especially early in design

Disposable



If you cannot afford to throw it away,
then it is not a sketch

But they are not "worthless"

Plentiful

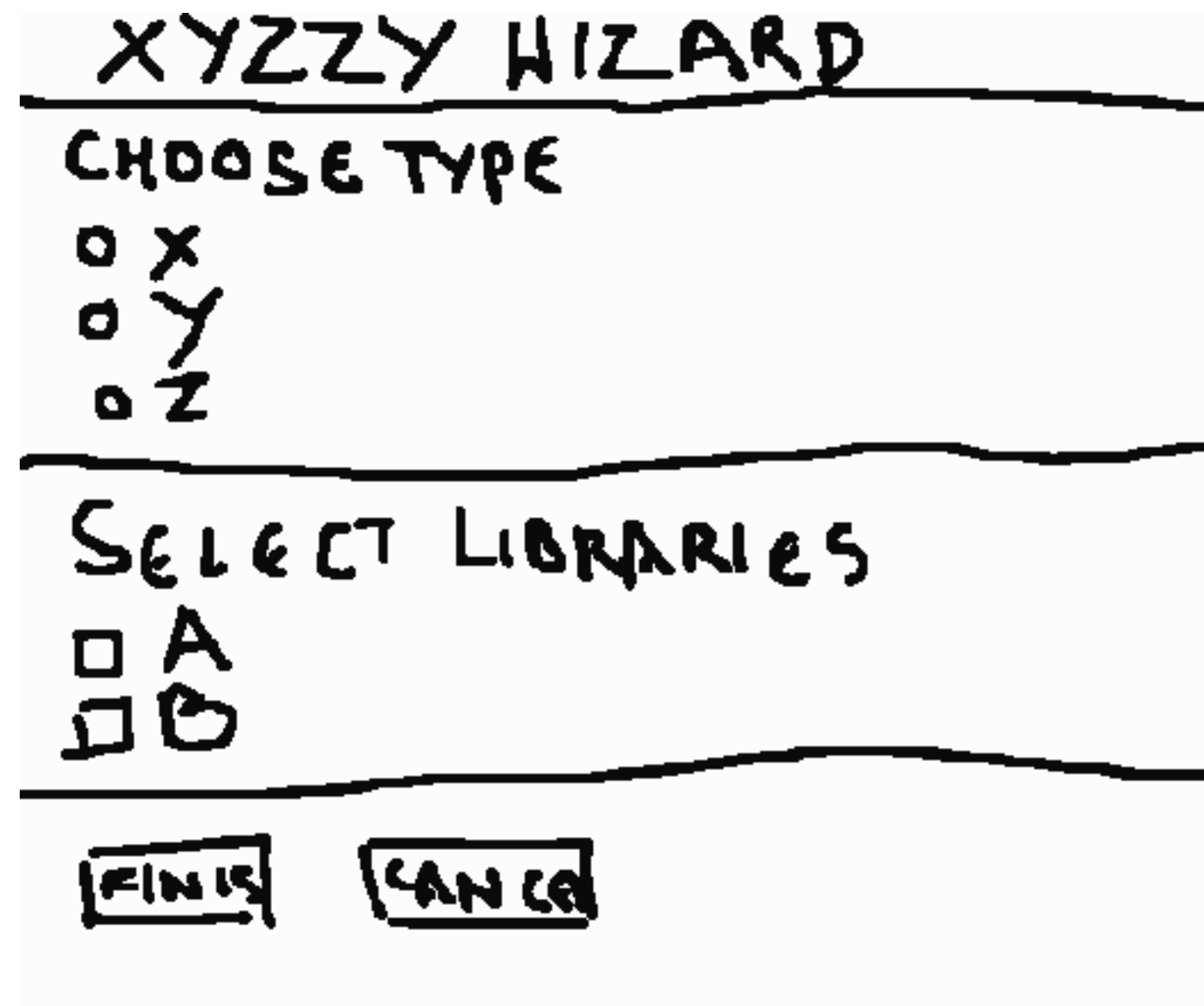


Sketches do not exist in isolation

Sketches are made to be compared

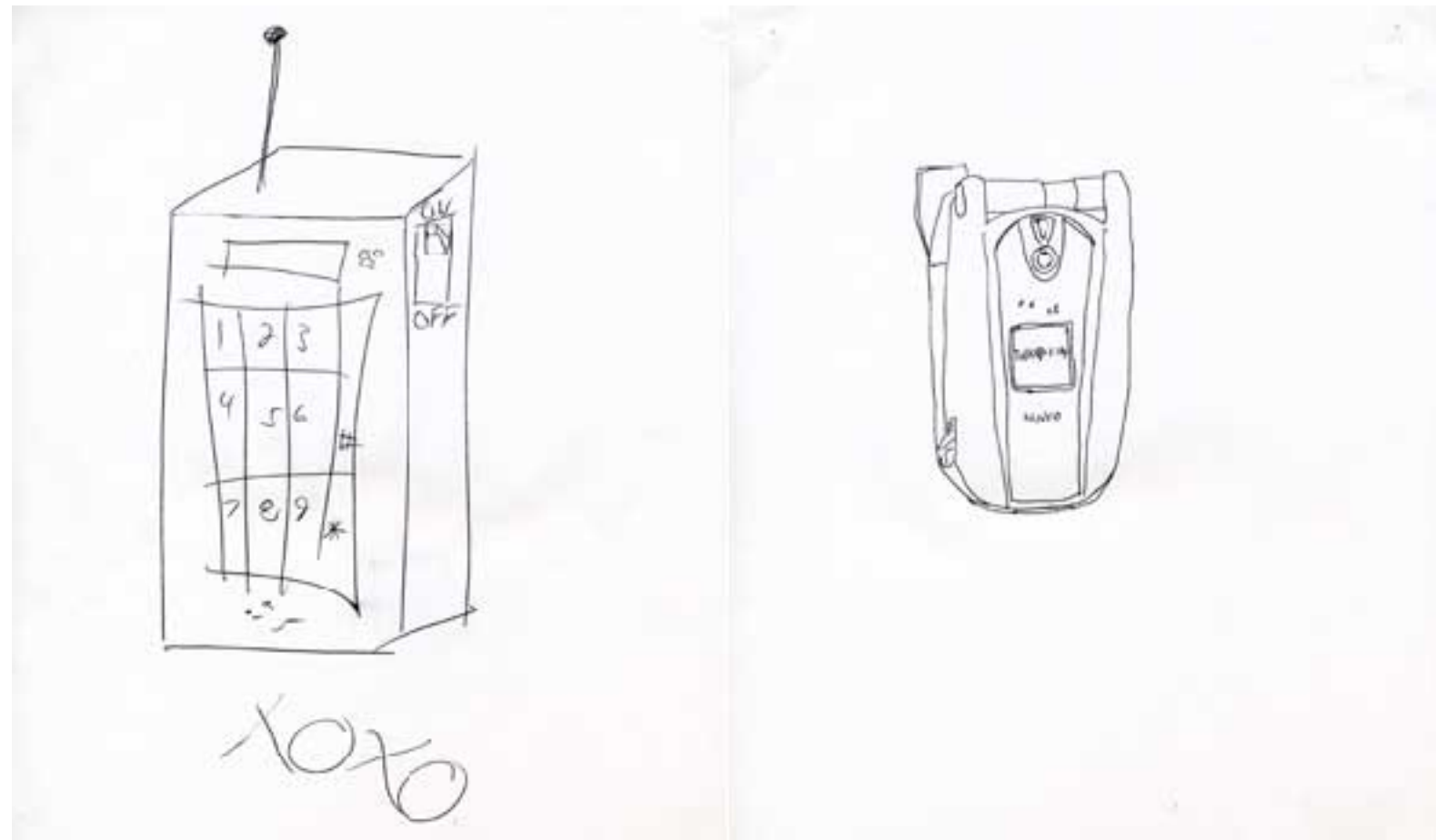
Meaning and relevance
is in the context of a
collection or series

Clear Vocabulary



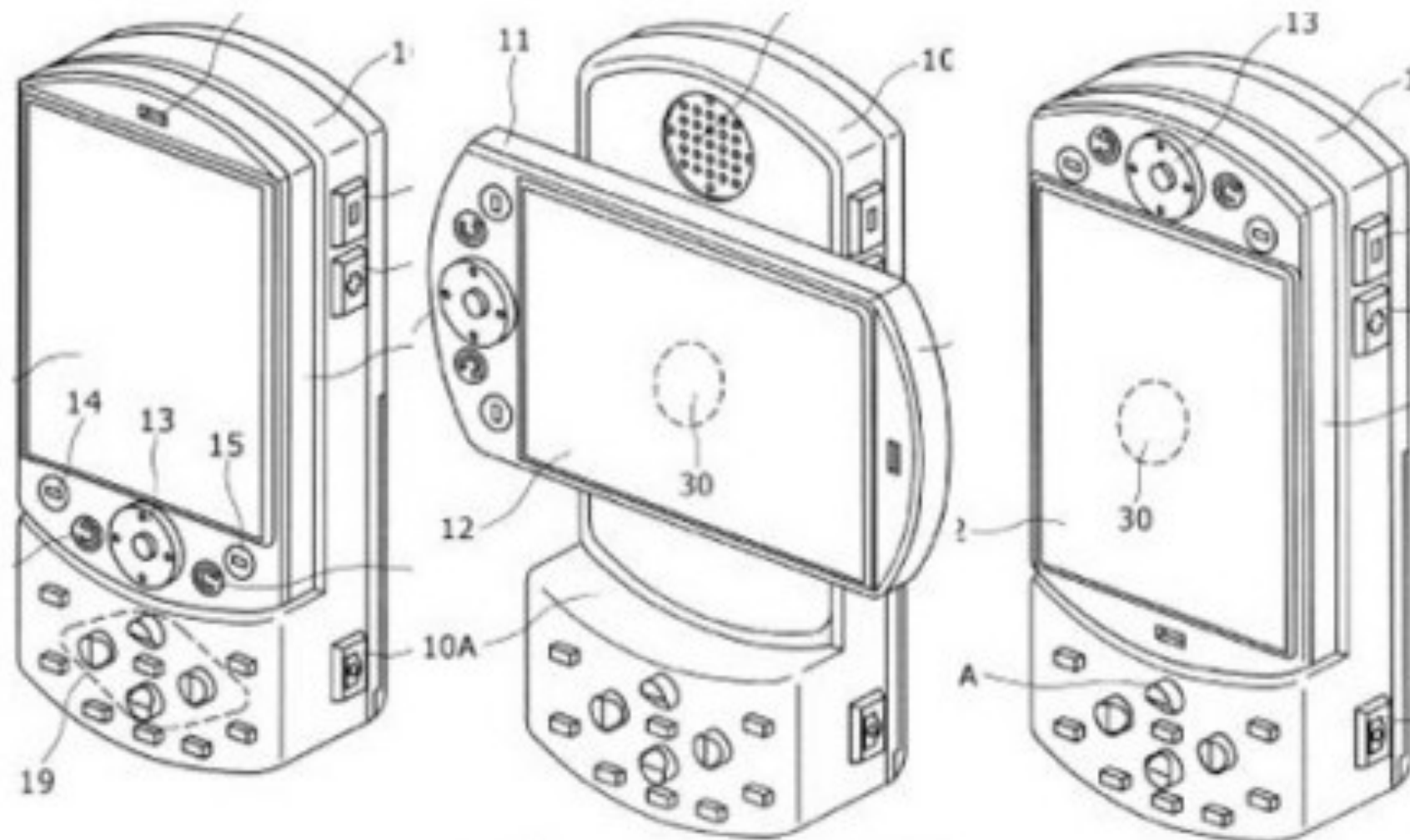
The way it is rendered makes it distinctive that it is a sketch (e.g., style, form)

Distinct Gesture



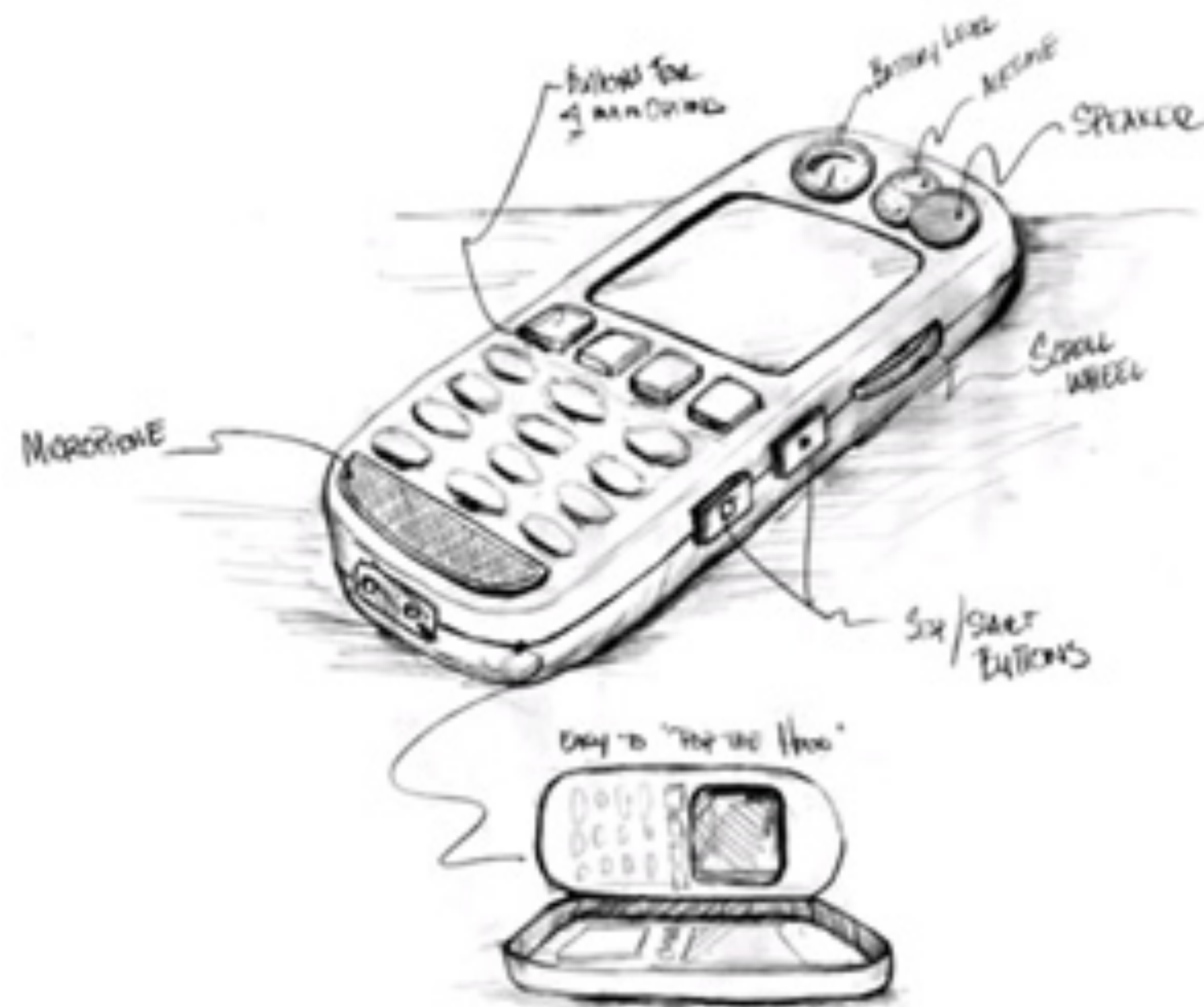
Fluidity of sketches gives them a sense of openness and freedom

Opposite of engineering drawing, which is tight and precise



Minimal Detail

Include only what is required to render the intended purpose or concept



Create JSP for this page

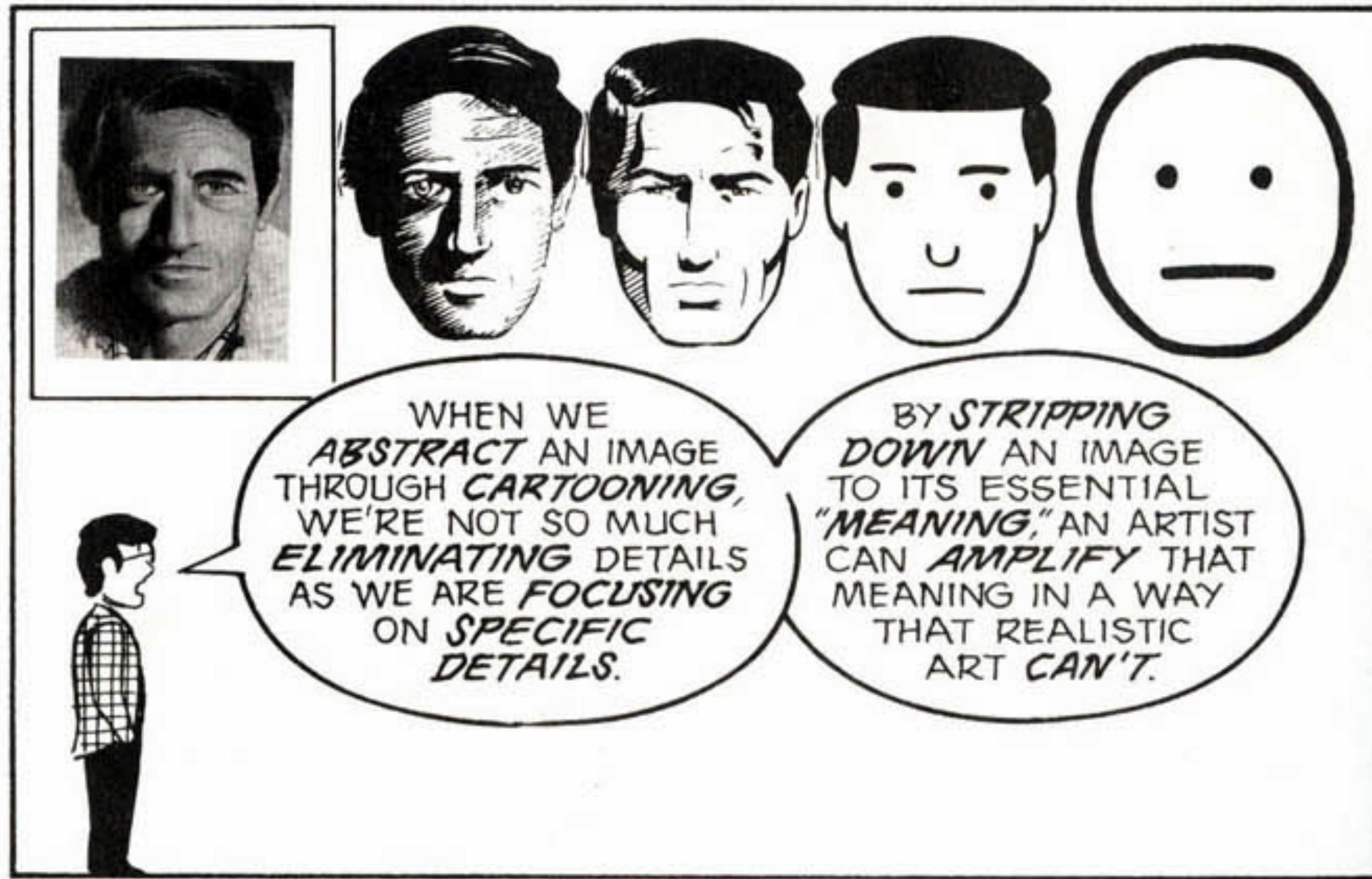
Name:

Number:

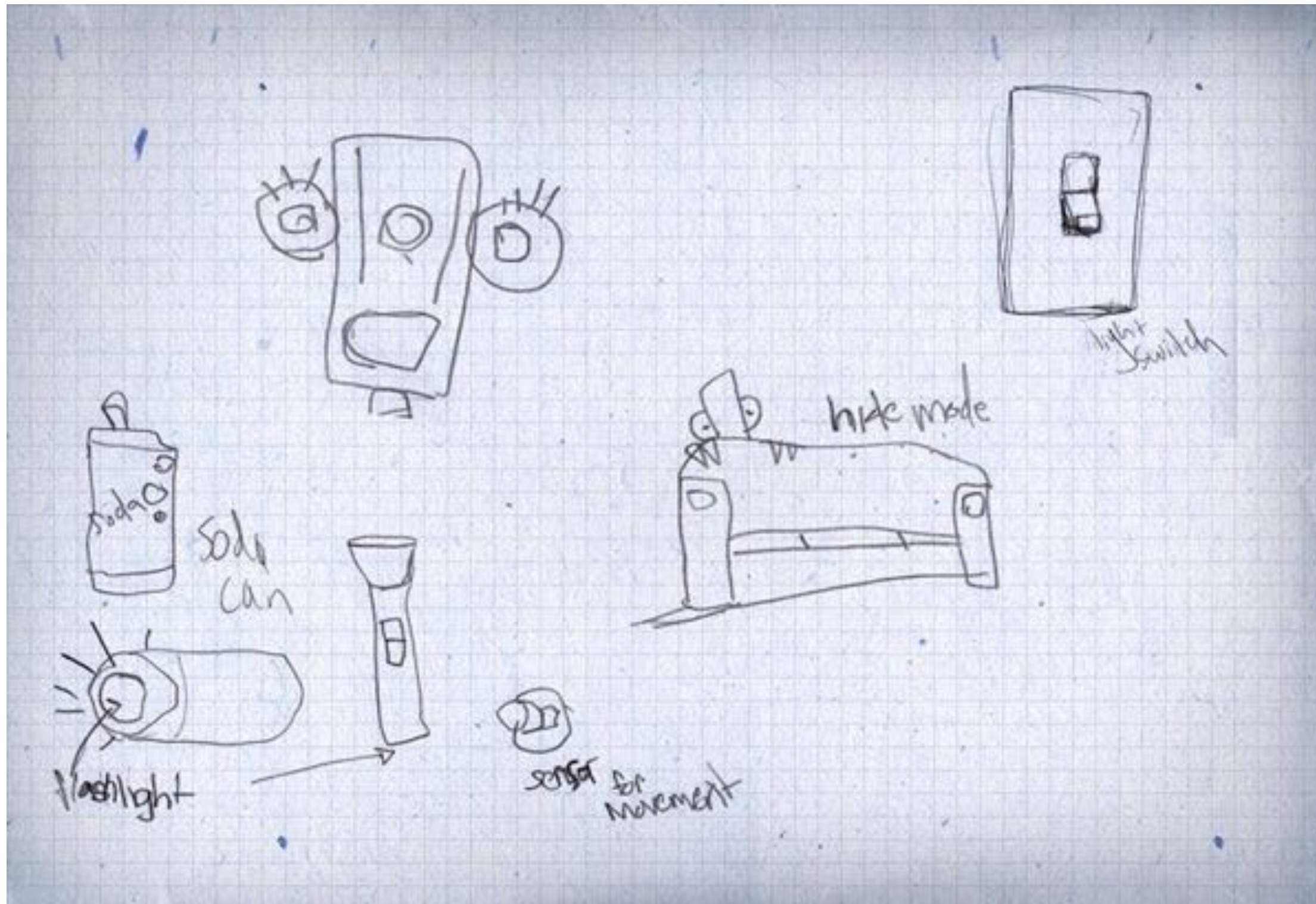
Category:

Price Range: to

Minimal Detail



Appropriate Degree of Refinement



Make the sketch as refined as the idea

If you have a solid idea,
make the sketch look
more defined

If you have a hazy idea,
make the sketch look
rougher and less defined

Suggest and Explore Rather than Confirm



Sketch should act as a catalyst to the desired and appropriate behaviors, conversations, and interactions with others (such as the people giving you feedback on your sketch).

Ambiguity



Intentionally ambiguous

Value comes from being
able to be interpreted
in different ways, even by
the person who created them

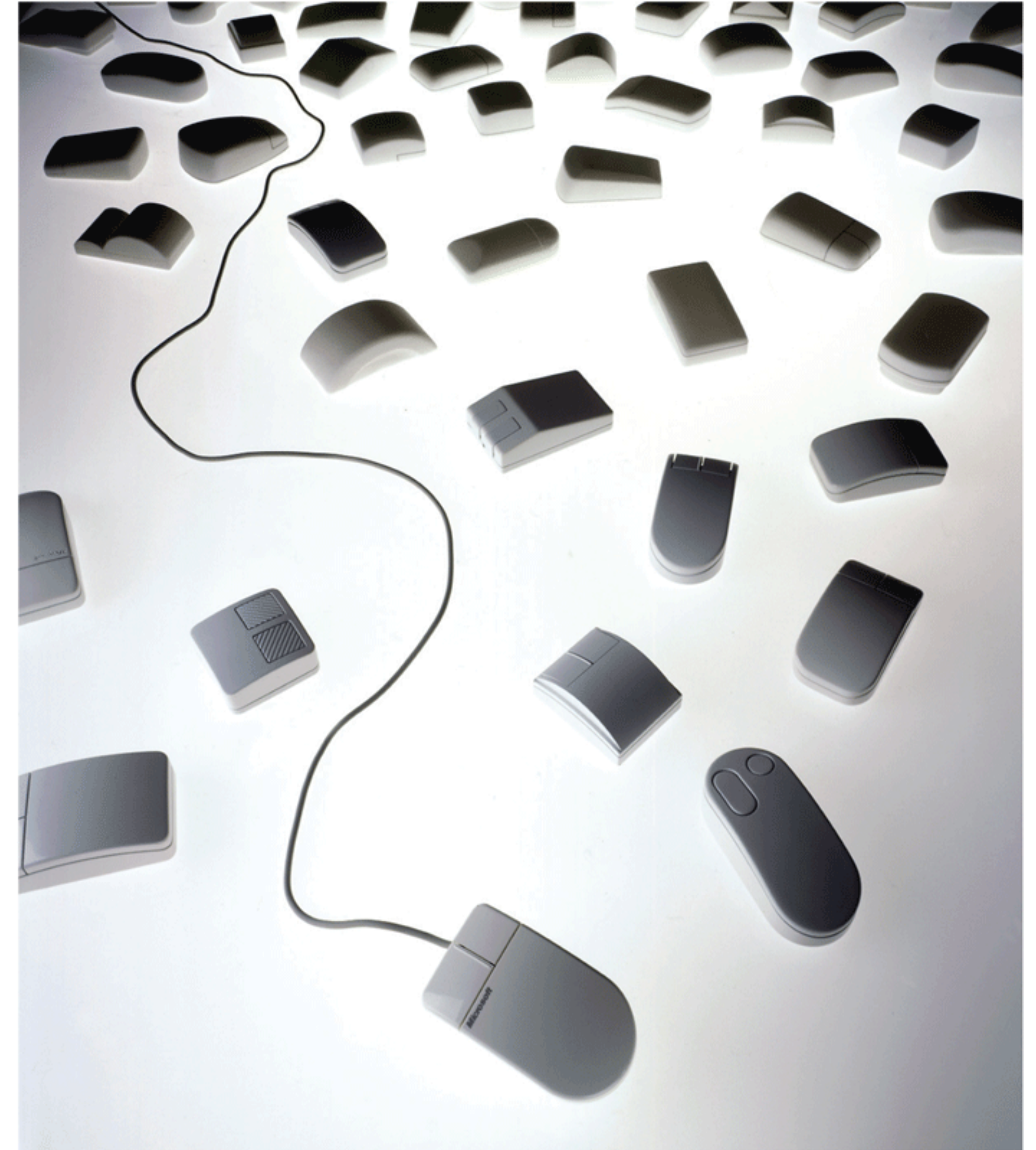
Sketches have holes

Rapid Prototyping

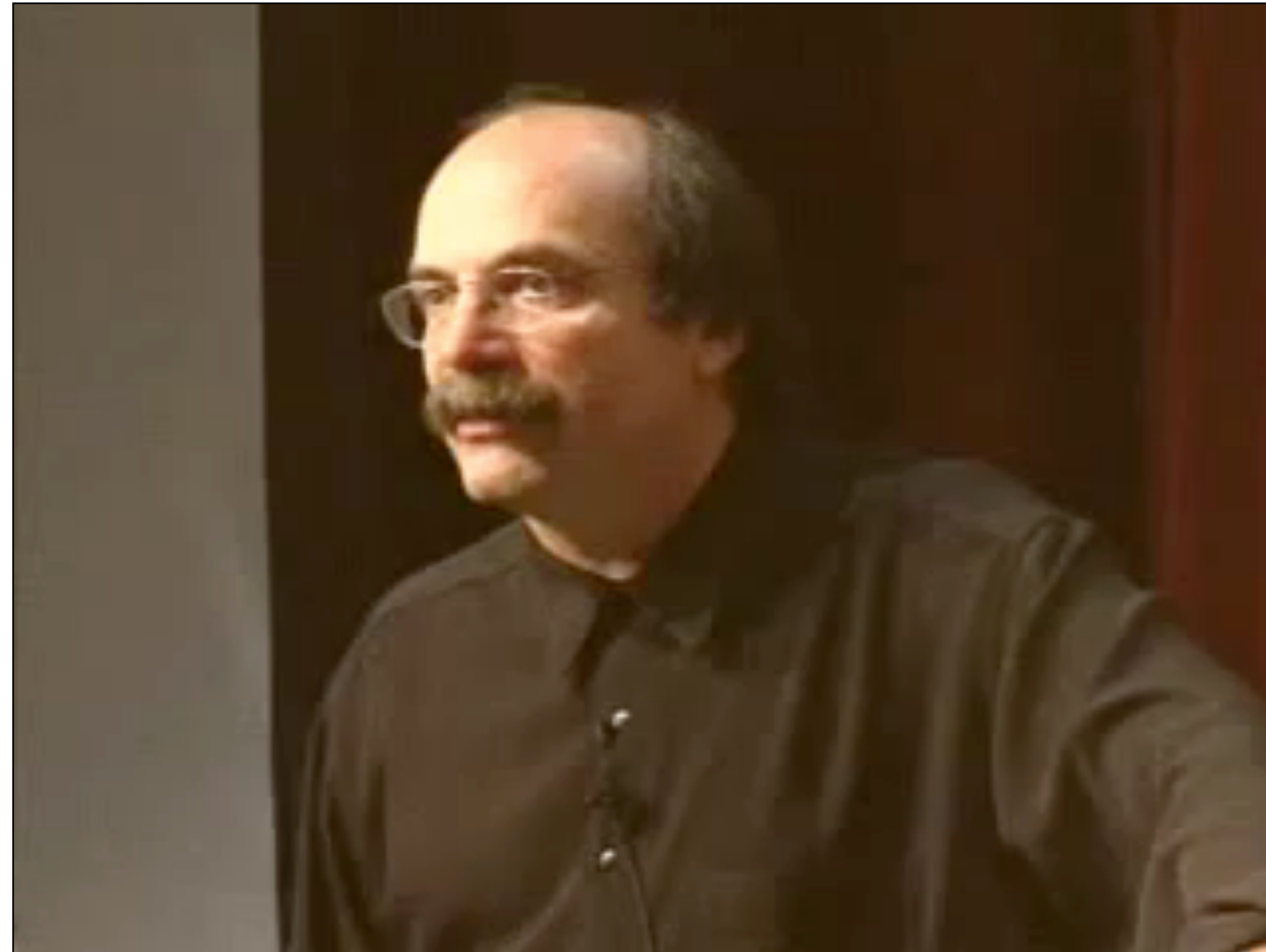
Moving from Sketches to Prototypes

Sketch	Prototype
Invite	Attend
Suggest	Describe
Explore	Refine
Question	Answer
Propose	Test
Provoke	Resolve
Tentative, non committal	Specific Depiction

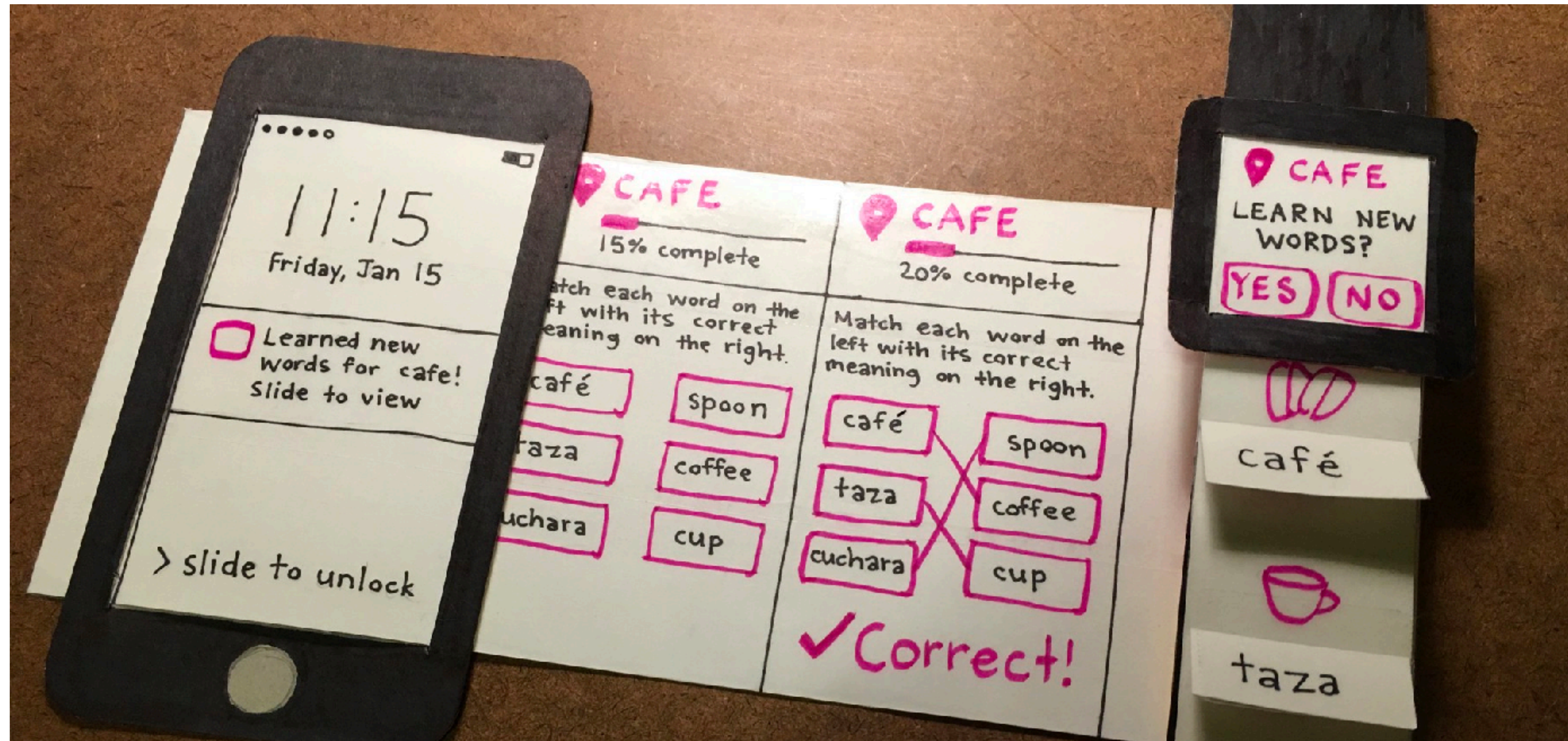
Example: Rapid prototyping the mouse



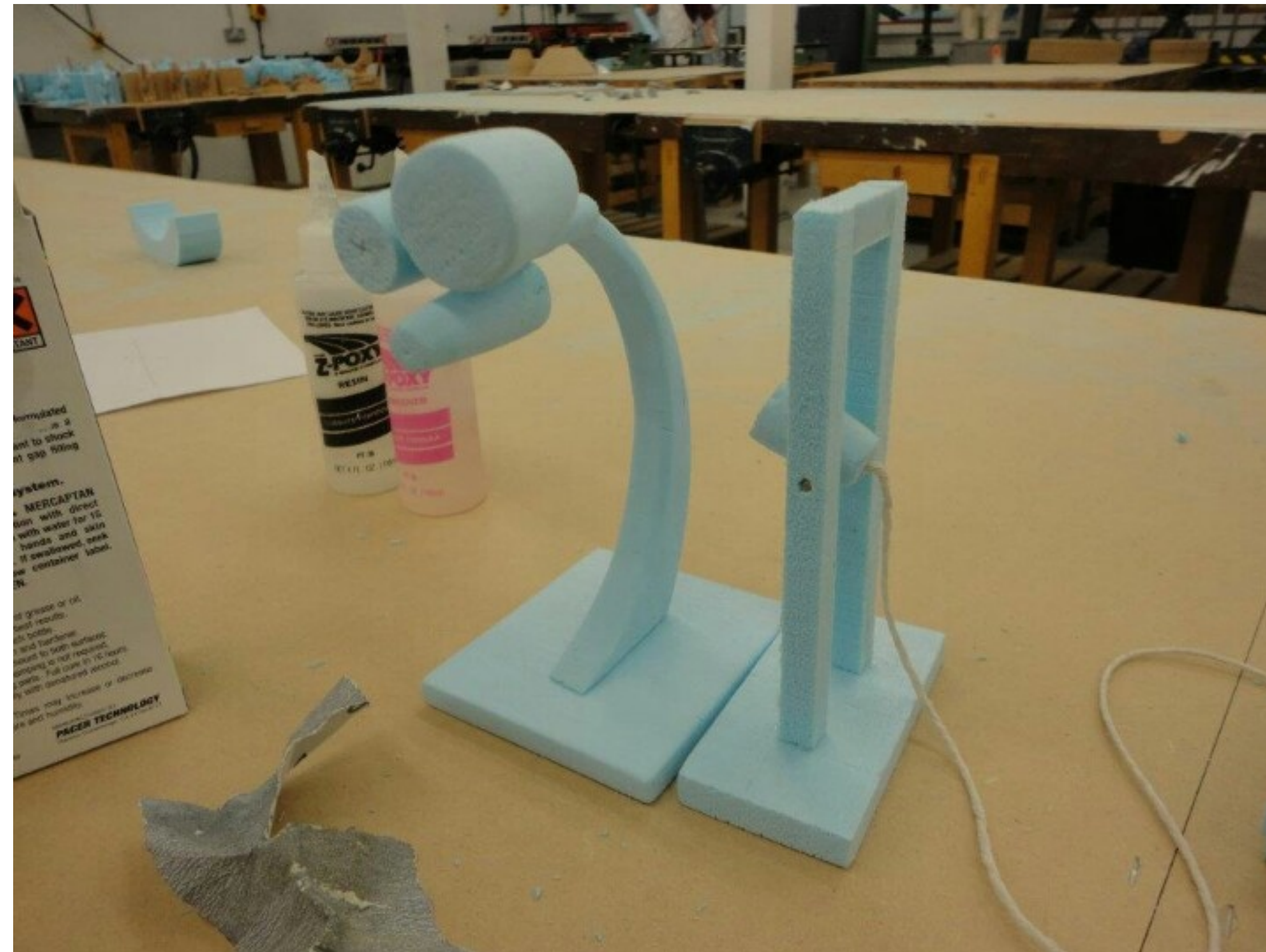
Example: Rapid prototyping the mouse



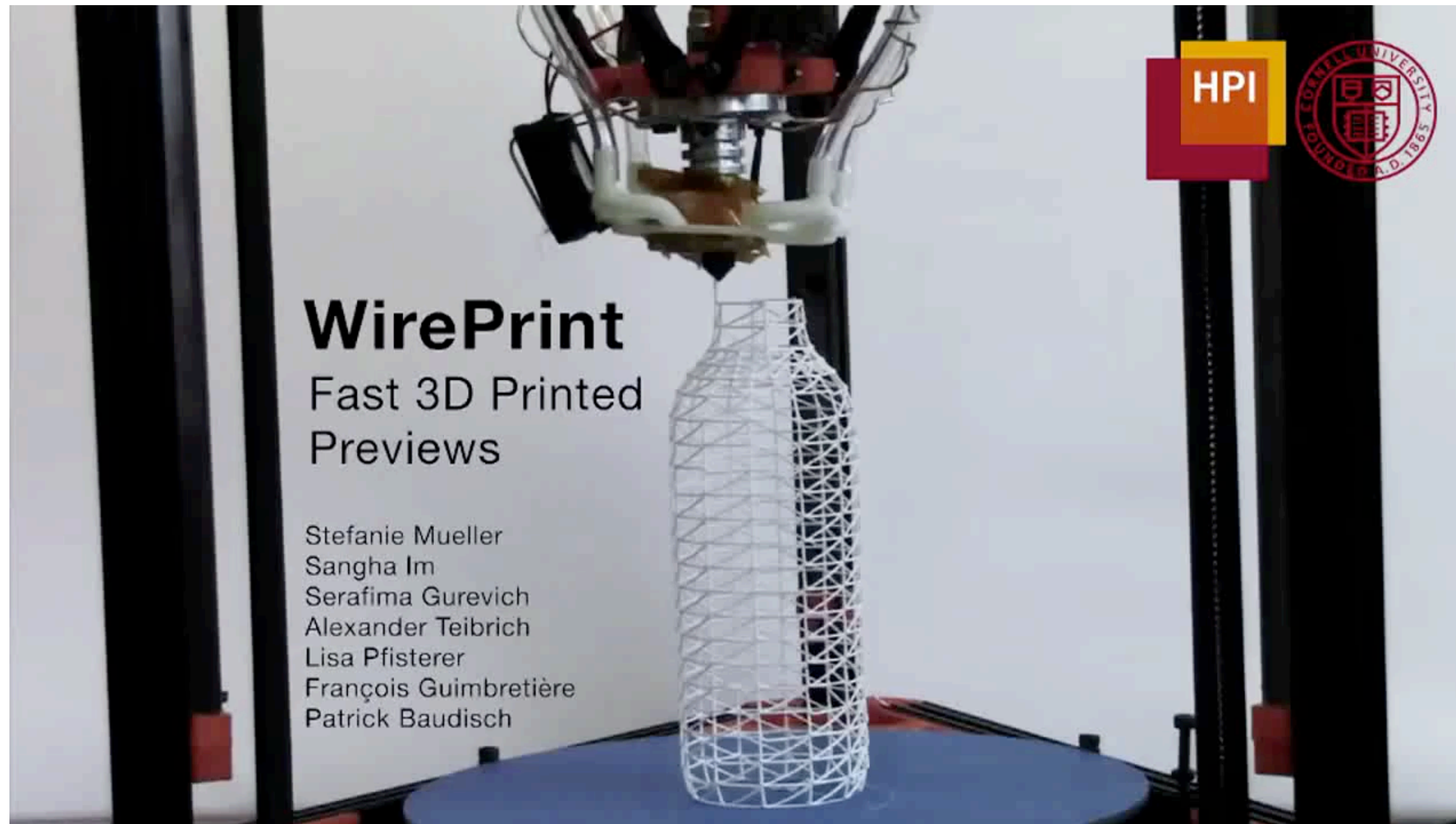
Paper Prototyping



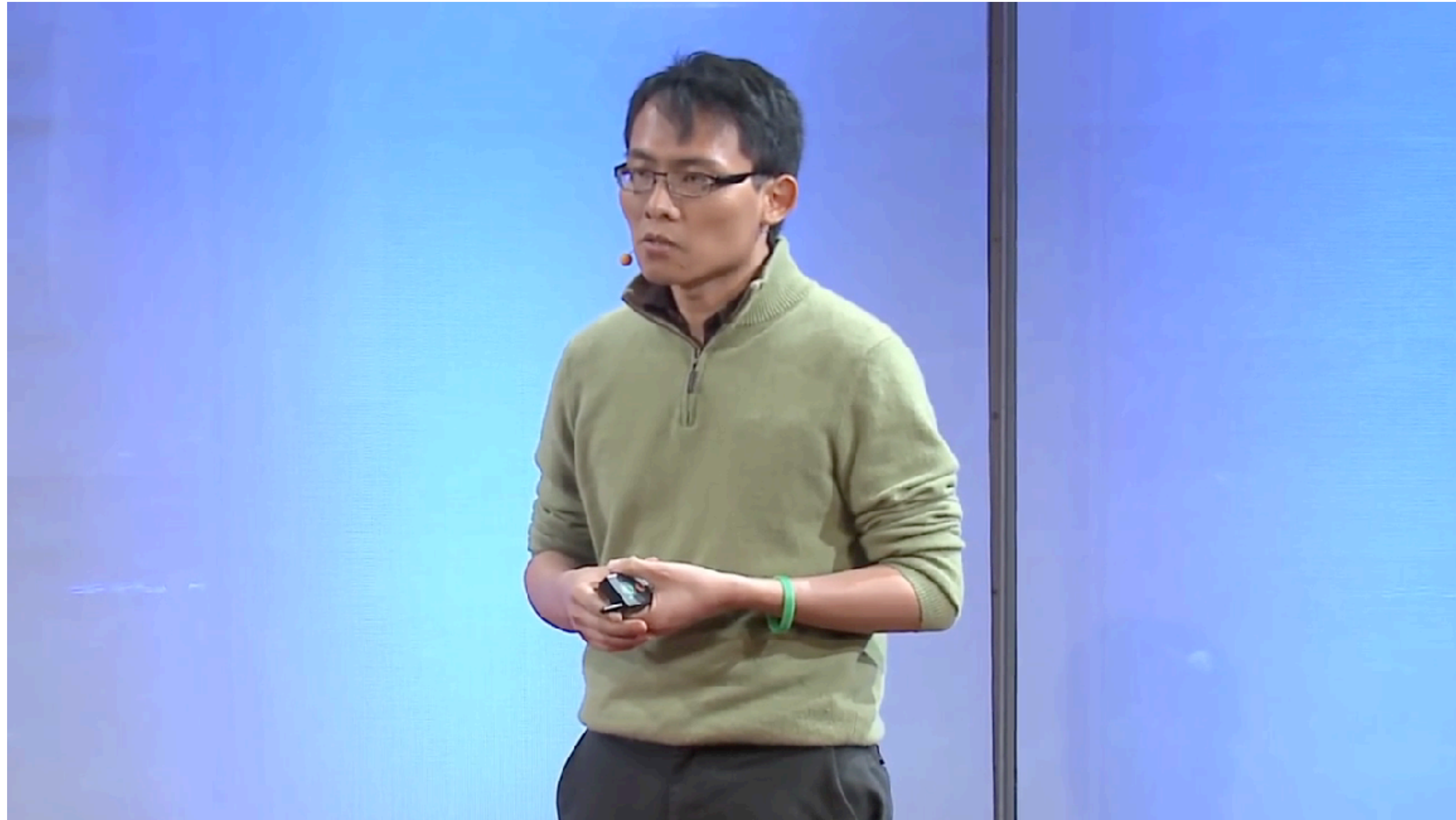
Rapid Physical Prototyping



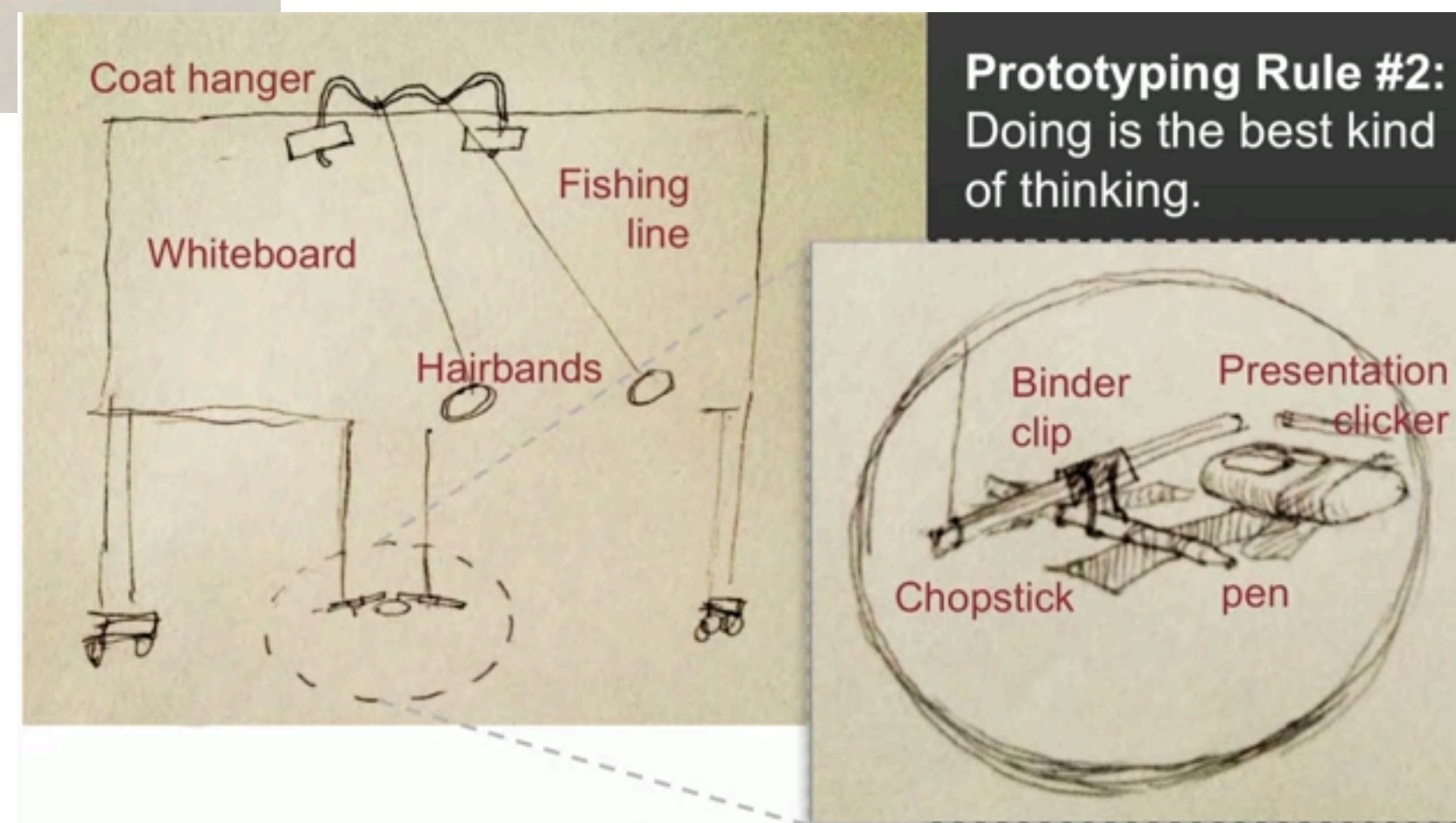
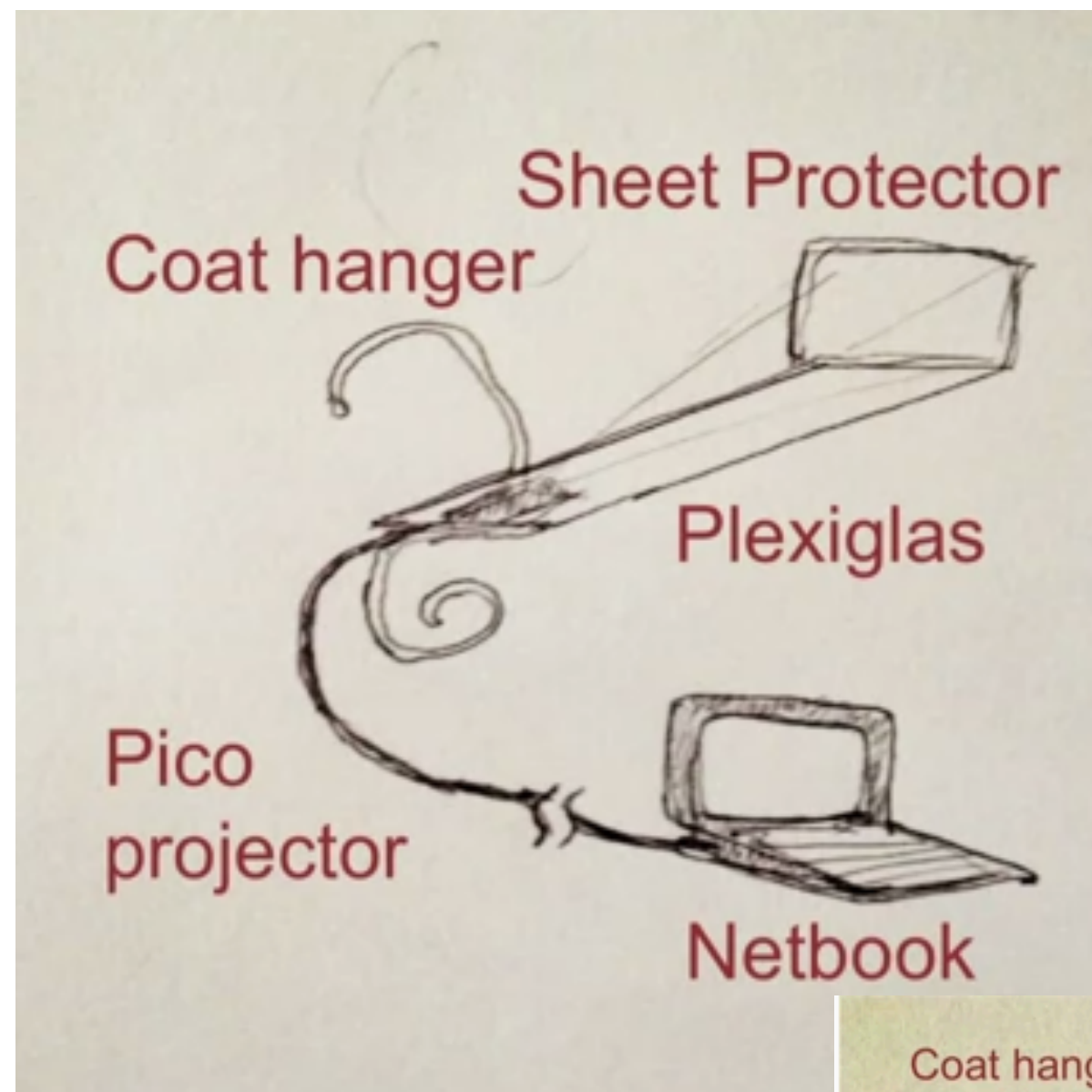
Rapid Physical Prototyping



Example: Rapid Prototyping Google Glass



Example: Rapid Prototyping Google Glass



Prototyping Rule #2:
Doing is the best kind
of thinking.